Venkata Lakshmi Parimala Pasupuleti

Fraud Data Analyst · Income & Subsidy Fraud Detection · Python · SQL · Anomaly Detection · Compliance Analytics

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PROFESSIONAL SUMMARY

Fraud Data Analyst with 7+ years of hands-on **income and subsidy fraud detection** in regulated affordable housing — reviewing income documentation, catching fake pay stubs, detecting household-composition discrepancies, and verifying asset disclosures across Section 8/HCV, LIHTC, and HUD portfolios at Entrata. Encoded that manual fraud-detection logic into automated compliance pipelines covering **millions of tenant certifications** across 100K+ unit portfolios, eliminating **80% of review workload with zero violations**. Built **Leasel**: an agentic AI pipeline that automates the income eligibility and fraud-flag determinations I ran manually — **93.6% precision, 87.9% F1**, every determination traceable to its triggering rule. Applied **Isolation Forest anomaly detection** across 3 research pipelines at GMU surfacing outliers manual review missed. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata
Jan 2021 – Aug 2024

Senior Analytic Software Engineer — Fraud Detection Automation · Compliance Analytics · Data Infrastructure

- Eliminated **80% of manual income fraud review workload** with zero compliance violations by encoding 7 years of manual fraud-detection judgment into automated Section 8/HCV, LIHTC, and HUD pipelines — AMI threshold calculations, income misrepresentation flags, recertification workflows, and escalation routing across **millions of tenant certifications**.
- Migrated all client data to **AWS S3** with Libsodium-encrypted pipelines — **100% PHI/HIPAA compliance**, zero security incidents; built fraud-aware data quality gates on Entrata's first production AI pipeline across **1.6M real financial records** (Python, PyTorch, OpenAI API).
- Optimized **200+ SQL queries for 30% performance improvement** across 1M+ property records; built property performance dashboards tracking eligibility rates, delinquency, rent-vs-AMI trends, and waitlist demand across 100K+ unit portfolios at **99.5% uptime**.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Income Fraud Detection · Eligibility Verification · Compliance Pipelines

- Reviewed and verified income documentation for **millions of tenant certifications** — identifying income misrepresentation (understated income to qualify for LIHTC caps, overstated for market-rate ratios), fake pay stubs, household-composition discrepancies, and concealed assets within fair-housing constraints requiring uniform document-driven verification.
- Applied **LIHTC/Section 42 and HUD income-limit rules** to determine household eligibility across 100K+ units and multiple jurisdictions — flagged and escalated misrepresented applications; verified asset disclosures against eligibility thresholds for affordable-housing subsidy integrity.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Compliance Reporting · SQL · Eligibility Systems

- Built compliance reporting features and SQL data layers supporting Section 8/HCV and LIHTC eligibility workflows; wrote and tuned queries across all client database environments; applied income-limit rules in certification tooling across 100K+ unit portfolios.

George Mason University
Aug 2024 – Aug 2025

Research Associate — Anomaly Detection · Statistical Modeling · Data Analytics

- Applied **Isolation Forest anomaly detection** across 3 research pipelines — surfaced statistically significant outliers manual review missed in public health and road safety datasets; built regression models achieving **$R^2 = 0.977$** across 190+ countries (Python, R) with automated reporting.
- Designed self-service BI dashboards (Tableau, ggplot2, Quarto) for faculty leadership — improved research comprehension **40%** across two cohorts; applied hypothesis testing, cross-validation, and A/B statistical techniques across multiple research projects.

PROJECTS

Leasel: Automated Income Fraud & Compliance Audit Pipeline
AI · Compliance

Python · LangGraph · RAG · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Automated the income eligibility and fraud-flag determinations I ran manually at Entrata: an agentic LangGraph pipeline extracts income, household, and asset data from **1,240 lease PDFs**; a deterministic rules engine applies LIHTC/Section 42 and HUD limits to flag violations — compliance logic kept deterministic because these determinations must be exact and fully auditable.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules with ~40% injected violation rate — pass@1 81.7%, pass@3 88.3% via multi-trial eval harness with regression gate; every determination traceable to its triggering rule with full metadata lineage.

CrashClarity: Large-Scale Risk Classification (104K Records)
ML · Risk

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and temporal risk variables; achieved **85%+ classification accuracy**; deployed interactive self-service risk dashboard on GitHub Pages.

AgentProbe: Multi-Agent AI Testing & Eval Toolkit
GenAI · Evals

Python · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) and 15-endpoint FastAPI backend — CI/CD integrated so every commit is automatically evaluated against compliance and quality thresholds.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML pipelines, anomaly detection, data governance, AI systems

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Fraud & Risk Analytics: Income Misrepresentation Detection, Household Composition Fraud, Asset Concealment Verification, Subsidy Fraud Patterns, Isolation Forest (Anomaly Detection), Rule-Based Fraud Engines, Deterministic Audit Pipelines, Escalation Routing

Analytics & ML: Python, R, SQL, Statistical Modeling, Random Forest, XGBoost, Logistic Regression, Feature Engineering, Cross-Validation, Hypothesis Testing, A/B Testing, Scikit-learn, Pandas, NumPy

Data Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), ETL Pipelines, PostgreSQL, MySQL, Oracle, SQLite, PHI/HIPAA Compliance, Metadata & Lineage, Docker, CI/CD (GitHub Actions), Git, Linux, PL/SQL

BI & Visualization: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Self-Service Dashboards, KPI Design, Data Governance

AI & NLP: LangGraph, RAG Pipelines, LLM-as-Judge Eval, Prompt Engineering, OpenAI API, Anthropic API, FastAPI, spaCy, BERT, Embeddings, LIHTC/HCV/HUD Regulatory Workflows